

Data paper

Character-Level Annotated Corpus of Four Chinese Dialects: Beijing, Dongbei, Shandong and Cantonese

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Author Roles

Huang Xinyu

- Data Curation: Collected and annotated Beijing dialect videos, edited video clips
- Annotation: Performed POS tagging and English glossing for Beijing dialect
- Writing - Original Draft: Authored README.md documentation
- Project Administration: Coordinated task deadlines

Luo Xiangyu

- Data Curation: Collected and annotated Dongbei dialect videos, edited video clips
- Annotation: Performed POS tagging and English glossing for Dongbei dialect
- Visualization: Guided video editing workflows (Audacity)

Ng Wang Shumei

- Data Curation: Collected and annotated Cantonese dialect videos, edited video clips

- Annotation: Performed POS tagging and English glossing for Shandong dialect
- Software: Developed Python scripts for metadata extraction

Shang Zichen

- Data Curation: Collected and annotated Shandong dialect videos, edited video clips
- Annotation: Performed POS tagging and English glossing for Cantonese
- Supervision: Oversaw annotation consistency across all dialects

Reuse Potential: Since our research project collects various Chinese dialects, including the Shandong, Beijing, Dongbei and Cantonese dialects. This project can be used for linguistic research to explore the common features and differences among dialects, such as lexical, phonology and syntactic variations. The transcribed and annotated audio data can be used to train automatic speech models specific to each dialect. Moreover, it provides the data for the machine translation of dialect. Furthermore, this dataset can be used for dialect learning courses and language preservation since it consists of authentic audio data recorded or produced by native speakers.

Abstract

This dataset presents 1 hours of audio recordings covering four major Chinese dialects (Beijing, Dongbei, Shandong, Cantonese) with parallel Mandarin translations and character-level annotations. Each utterance is segmented and tagged with English glosses and Universal POS tags, enabling cross-dialect linguistic analysis. Data was collected from naturalistic scenarios (family conversations, market transactions) using Zoom H6 recorders, then manually validated by native speakers. Repository location: The dataset is hosted on GitHub: <https://github.com/icey0211/language-study> License: CC-BY-NC 4.0 Potential reuse includes dialect preservation studies and multilingual ASR model training.

Keywords: Chinese dialects; Pos tagging; Computational linguistics; Language preservation; Multilingual corpus; Open data

(1) Overview

Repository location:

The dataset is openly available on **GitHub**: <https://github.com/icey0211/language-study>

Context

This dataset was created as part of a course project for English study and digital communication/A4B093 at The Education University of Hong Kong (EdUHK).

It aims to:

- Document linguistic variations across four major Chinese dialects (Beijing, Dongbei, Shandong, Cantonese)
- Provide fine-grained annotations (English + POS tags + parallel Mandarin translations) for computational analysis
- Serve as a benchmark for dialect preservation

Key Features Summary

Aspect	Description
Data Source	1hr videos/voice recording + manual verification
Annotation	Character-level English gloss & Universal POS tags (UPOS)
Size	1 hours of audio (15min per dialect) → 815 annotated sentences
License	CC-BY-NC 4.0(Free for academic use, non-commercial)

Why This Dataset?

1. Unprecedented Annotation Granularity

The first publicly available corpus providing:

1. Character-level English glossing (e.g., Cantonese "咗" → "[perfective aspect]")
2. Parallel Mandarin translations with POS tags (UPOS standards)

3. Enables precise linguistic analysis (e.g. tracking Shandong dialect grammaticalization)

2. Geolinguistic Diversity

Covers 4 major dialect groups:

1. Northern Mandarin: Beijing, Dongbei
2. Shandong
3. Southern Yue: Cantonese (Hong Kong SAR standard)

3. AI-Ready Structured Data

CSV formats with:

1. Timestamped audio clips
2. Machine-readable metadata (speaker demographics, recording context)

(2) Method

1. Video Collection:

Searched for authentic dialect videos on YouTube and Bilibili using keywords like "北山东方言 山东话" or "粤语".

2. Voice recording: Record a certain duration of dialect by oneself as the subject for research and analysis.

Developed a Python script to download videos.

3. Audio Processing:

Extracted audio using python, then transcribed with it.

Use iFlytek's simultaneous interpretation feature to transcribe the text, then supervise and manually correct errors made by the AI

Manually corrected transcription errors by comparing with the native speakers per dialect. (as AI and Python-based transcription may lack full accuracy).

3. Segmentation:

Imported WAV files into Audacity to clip into sentence-level units (2-14 seconds each).
Packaged clips into ZIP archives named by dialect and the number of WAV files (e.g., Beijing_Story_1-50-HXY-Female-21_20250412.zip).

4. Linguistic Annotation:

Split sentences into characters and annotated each character with:

English gloss (e.g., 咗 → "[perfective aspect]")

Universal POS tag

Mandarin equivalence (e.g., 俺 → 我)

(3) Dataset Description

Repository name – GitHub

Object name – Beijing, Dongbei, Shandong and Cantonese Dialect Contrast Corpus

Format names and versions – CSV format, WAV (audio files)

Creation dates – Start: 2025-04-10

End: 2025-04-20

Dataset creators

Huang Xinyu - Data collection, Annotation, Dialect Transcription, Audio Cutting

Luo Xiangyu - Data collection, Annotation, Dialect Transcription, Audio Cutting

Shang Zichen - Data collection, Annotation, Dialect Transcription, Audio Cutting

NgWang Shumei - Data collection, Annotation, Dialect Transcription, Audio Cutting

Language

- Primary: Beijing, Dongbei, Shandong and Cantonese Dialect.
- Metadata/Annotations: English, Chinese.

License – This dataset is licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, which means that anyone can share, change, and build on it for any reason, even for profit, as long as they thank the original authors.

Publication date – 2025-04-20

(4) Reuse Potential – Researchers in linguistics, artificial intelligence, and cultural studies can use this information in a lot of different ways. The carefully labeled dialectal corpus has Beijing, Dongbei, Shandong and Cantonese Dialect varieties, along with English glosses and POS tagging. It can be used for many things, from preserving dialects and comparing languages to teaching advanced speech recognition and machine translation systems. Researchers could combine this data with other resources for regional languages to test linguistic theories or make materials to help people learn dialects. Including both textual and audio identifiers helps the progress of speech technology even more. However, the current coverage of regional dialects, the fact that colloquial expression notes may not always be accurate, and the need for extra audio files may make reuse more difficult. For more people to use it, it will be important to have clear details of the annotation standards and licensing terms. The dataset is very useful for studies that combine language documentation with AI innovation because it has a unique mix of linguistic information and computer readiness.

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Competing interests

The authors declare no competing interests, financial or non-financial, related to this study.

Data collection was conducted with participants' informed consent, and all audio materials were anonymized to protect speaker privacy.

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